COMP710: Assignment: Individual Game Project

ISSUED: WEEK 5, WEIGHTING: 35%

ASSIGNMENT DEADLINES

Due date and time; deliverables submitted via Blackboard:

- First-Walk-and-Talk (FWT) Milestone: **Week 8, 11:59 pm, Monday**
- Gold Milestone: **Week 9, 11:59 pm, Monday**

INDIVIDUAL GAME PROJECT

Individually design and develop a real-time interactive video game in C++ and Visual Studio, utilising techniques and technologies presented in class. The game must showcase game programming aspects learnt during studio sessions and has the scope to include additional unfamiliar, and sometimes complex, game programming solutions.

The development is to utilise the following game production stages:

- Pre-production – Design Phase
- Production – Development Phase
- Post-production – Reflection Phase

All work undertaken is to be stored and displayed on GitHub, with regular commits of work-in-progress incrementally throughout the development cycle (pre-production, production and post- production).

PRE-PRODUCTION – DESIGN PHASE

Design a simple sprite-based game to implement using object-oriented programming techniques in C++, extending upon the COMP710 "GP Framework" as developed in class. The player must be able to win or lose the game. The scope of the design should be limited to ensure the project is achievable, however do not limit the scope such that it does not reflect an adequate amount of effort (nor should it be a clone of "Space Invaders", "Robotron", or "Asteroids").

Firstly, create a one-page Game Pitch document. This must be non-technical in nature and must clearly convey the core purpose of the game. The Game Pitch must cover the following topics:

- Game title/name;
- Genre;
- Target audience;
- Game play and mechanics;
- Player goals;
- Key/core features;
- User interface design;
- Unique selling points.

The Game Pitch document may also have one supporting image embedded in the document, if required.

Once you are satisfied with your Game Pitch, create a Game Design Document (GDD) which describes how the game will be played. This document must convey the vision for the game

and will be a multipage document containing comprehensive design details, and will also include a title page,

table of contents and bibliography (in APA 7th Edition format). Choose appropriate sections and headings for your game design, but ensure the GDD includes coverage on the following topics:

- Game rules, mechanics and specific feature decisions for your design;
- Key algorithms which govern game play elements;
- Control scheme;
- Mock-up interface details, including sample screen layout for each stage of the game;
- Cheat features;
- A required asset list, including but not limited to: sprites, animations, and audio.

Next, create a Technical Design Document (TDD) which describes how the game will be developed. Analyse, design, and document technical solutions for in-game scenarios. Select, adapt and apply a variety of game-related simulation techniques, data structures and algorithms for implementation in game. The TDD will be a multipage document that comprehensively describes technical design detail and planning, and will also include a title page, table of contents and bibliography (in APA 7th Edition format). The TDD must include coverage on the following topics:

- Descriptions of logic and technical algorithms required to implement game features;
- Flowchart, UML Activity, Sequence or State diagrams which illustrate the technical design of important systems to be implemented;
- UML Class diagram(s) which detail the object hierarchy, object interaction, and key properties and functionality to be implemented;
- Debug features;
- Coding standards and naming schemes;
- Relevant file formats;
- An acceptance test plan which features ten questions regarding game functionality. By answering all ten questions, a tester could validate that the game has been completely developed.

PRODUCTION – DEVELOPMENT PHASE

Implement the game based upon the Game Design and Technical Design Documents. The Game Design and Technical Design can evolve as development progresses. If there are aspects of the game that have been poorly considered during the Design Phase, and additional design must be undertaken during Production, this is okay. If there are aspects of the game that can be improved by implementing features differently to initially designed, go ahead and implement the improved vision. Ensure regular and well-commented commits to GitHub occur throughout the Development Phase. Two milestone deliveries will occur via Canvas, a *First-Walk-and-Talk* milestone, and the final *Gold* milestone.

POST-PRODUCTION – REFLECTION PHASE

Once development is complete, create two final documents, a **`readme.pdf`** and a post-mortem. The **`readme.pdf`** must contain all authorship details as well as game controls. The post-mortem document must report on the game development process by reflecting upon the design and development processes undertaken. It should describe areas of difficulty, and the problem-solving methodologies applied to resolve issues in design and development. The document must be at least 1250 words in length, and clearly include the following five

sections: Overview, What Went Right, What Went Wrong, Lessons Learnt, and Conclusion.

TECHNICAL GAME FEATURES

The game is required to demonstrate the following features:

- Appropriate, effective and correct usage of C++ and object-oriented programming;
- Visual Studio Solution, Project and Source/Header files;
- Real-time interactive game programming simulation techniques;
- 2D Graphical effects: sprites, animated sprites, scaling, rotation, layering, particle systems, text, alpha and colour blending;
- Xbox 360 game controller(s);
- Audio;
- Artificial intelligence;
- Data driven design;
- Game programming middleware and libraries;
- Testing and debugging features, including audit trail logging.

MIDDLEWARE AND THIRD-PARTY RESOURCES

The following middleware and libraries are approved for use in the assignment:

- COMP710 "GP Framework";
- C++ Standard Template Library (STL);
- Simple DirectMedia Layer (SDL2) including SDL2_Image and SDL2_TTF APIs;
- The OpenGL Extension Wrangler Library (GLEW);
- Dear ImGui;
- FMOD Core (Low-Level) API;
- Box2D.

Permission to utilise any other third-party, externally created assets must be sought from the content's author and the course lecturer. This includes art assets, sound assets, and source code. If students wish to use any third-party code or middleware this must be discussed with, and approved by, the course lecturer prior to the development phase. All requests and approvals must formally occur via AUT's Canvas *Discussions* forum.

INTERFACE FEATURES

The game executable must provide an intuitive interface with the following features:

- Fullscreen – the game runs in fullscreen mode by default;
- Splash screen(s) – the game starts with an AUT splash screen, and any other splash screens required by the middleware;
- Instructions – clear in-game game instructions, including input control summary/overview;
- Restart – after winning/losing, a new game can be started without exiting the executable.

BUILD AND RUNTIME QUALITY

Write clear and efficient game source code. The source code must compile at Warning Level 3, with no errors or warnings present. The *Gold* milestone executable must be free of memory leaks, runtime bugs and crashes.

FIRST-WALK-AND-TALK AND GOLD MILESTONE GITHUB LOGS

Create a text file containing a log of all your GitHub commits and associated messages.

MILESTONE – FIRST-WALK-AND-TALK

This is to be a work-in-progress build of the game that is playable and demonstrates key

features of the game. A release build executable must be included with the *First-Walk-and-Talk* submission and all assets required to run the game must be included. Ensure Visual Studio's project settings are set to Release when creating this build.

MILESTONE – GOLD

This is to be the final build of the game that would be "Gold Mastered" for release. A release build executable must be included with the *Gold* submission and all assets required to run the game must be included. Ensure Visual Studio's project settings are set to Release when creating this build.

SUBMISSIONS

For each submission, accurately complete the electronic Individual Assignment Coversheet PDF document and include this in the submission zip. The coversheet is available on Canvas in the Assessment section. Include a text file containing a copy of all your GitHub commits and associated messages. Include all source code, Project and Solution files and game assets, as well as Debug and Release build executables, in the submission zip. Submit the zip via Canvas, named as follows.

SUBMISSION – FIRST-WALK-AND-TALK

Deliverable Due: **Week 8, 11:59 pm, Tuesday**

Include the Game Pitch, Game Design and Technical Design Documents as PDF files in the First-Walk- and-Talk (FWT) submission zip. The required structure is as follows:

- **FWT - Student_ID.zip** First-Walk-and-Talk submission zip.
  - **Coversheet - Student_ID.pdf** AUT Individual Coversheet.
  - **docs** Developer references.
    - **Pitch - Student_ID.pdf** Game Pitch Document.
    - **GDD - Student_ID.pdf** Game Design Document.
    - **TDD - Student_ID.pdf** Technical Design Document.
    - **GHL – Student_ID.txt** GitHub Log document for FWT.
  - **game** First-Walk-and-Talk Build.
    - **codename.exe** Game release build exe.
    - **codename_debug.exe** Game debug build exe.
    - ***.*** All required asset files.
  - **lib** Any additional third-party libraries.
  - **source** Folder of source build files.
- **codename.sln** Visual Studio Solution.
  - ***.*** Layout dependant on game.
  - **test** Supporting files for testing.

SUBMISSION – GOLD

Deliverable Due: **Week 9, 11:59 pm, Tuesday**
Include the **readme.pdf** and Post-Mortem documents as PDF files in the Gold submission zip. The required structure is as follows:

- **Gold - Student_ID.zip** Gold submission zip.
  - **Coversheet - Student_ID.pdf** AUT Individual Coversheet.
  - **docs** Developer references.
    - **Pitch - Student_ID.pdf** Game Pitch Document.
    - **GDD - Student_ID.pdf** Game Design Document.
    - **TDD - Student_ID.pdf** Technical Design Document.
    - **GHL – Student_ID.txt** GitHub Log document for Gold.
    - **Post-mortem – Student_ID.pdf** Post-mortem document.
  - **game** Gold Build.
    - **codename.exe** Game release build exe.
    - **codename_debug.exe** Game debug build exe.
    - **readme.pdf** Details author and controls.
    - ***.*** All required asset files.
  - **lib** Additional libraries.
  - **source** Folder of source build files.
- **codename.sln** Visual Studio Solution.
  - ***.*** Layout dependant on game.
  - **test** Supporting files for testing.

LATE SUBMISSION

Please see the COMP710 *Course Guide* for the late submission policy.

MARKING CRITERIA

Criteria:	Weight:	A	B	C	D

Design: • **Game Pitch** • **Game Design Document (GDD)** • **Technical Design Document (TDD)**	**12.5%**	All of the following: • Game Pitch, GDD and TDD are well structured, comprehensive and clear. Each document details the game design and development plan; • Documents have been proofread and are free of spelling and grammatical errors.	All of the following: • Game Pitch, GDD and TDD have relevant topic coverage and planning details, however production may be hindered in some instances due to missing detail(s).	All of the following: • Submission before the milestone deadline; • Game Pitch, GDD and TDD are all present, each describing core game qualities, however production will be hindered due to missing significant details.	Any one of the following: • No coversheet included; • No Game Pitch, GDD, or TDD PDFs submitted; • Poor topic coverage in Game Pitch, GDD or TDD.

Gold Game Implementation: • **Game play** • **Interaction** • **Input** • **Game restarting** • **Bugs** • **Crashes**	**25%**	All of the following: • The Gold Game Implementation "B" criteria; • Game is bug-free; • Game is crash-free; • Game is memory leak-free.	All of the following: • The Gold Game Implementation "C" criteria; • The game can be won and/or lost; • Game can be restarted without closing the executable.	All of the following: • Submission before the milestone deadline; • Simple game play present; • User input via game controller present.	Any one of the following: • No Gold Submission zip; • No coversheet included; • Gold milestone game is unplayable.

Gold Graphics Implementation: • **Graphical effects** • **Sprites** • **Particles** • **Animation**	**25%**	All of the following: • The Gold Graphics Implementation "B" criteria; • Multiple animated sprites present; • Multiple particle special effects present; • Scaling, rotation, alpha and colour blending SFX present.	All of the following: • The Gold Graphics Implementation "C" criteria; • GUI present; • HUD present; • Single particle special effect present; • Splash screen present.	All of the following: • Submission before the milestone deadline; • Multiple sprites present; • Single animated sprite present.	Any one of the following: • No Gold Submission zip; • No coversheet included; • No graphics present in gold milestone executable.

Gold Milestone Build Quality: • **Compilation** • **Overall game** • **Technical features** • **Complexity** • **Middleware usage**	**25%**	All of the following: • The Gold Milestone Build Quality "B" criteria; • Artificial intelligence present; • Data driven design present; • Complex game scenarios developed; • Unfamiliar features developed.	All of the following: • The Gold Milestone Build Quality "C" criteria; • Gameplay scenarios developed; • Debugging features present; • A variety of middleware utilised; • Audio present.	All of the following: • Submission before the milestone deadline; • Game utilises technologies and techniques presented in class. • Real-time simulation techniques utilised. • Source code compiles with zero warnings.	Any one of the following: • No Gold Submission zip; • No coversheet included; • No Visual Studio Solution, Project and C++ source code present; • Source code does not compile.

Reporting: • **GitHub usage** • **First-Walk-and-Talk Milestone** • **Post-mortem**	**12.5%**	All of the following: • The Reporting "B" criteria; • Comprehensive, insightful and honest post-mortem; • Post-mortem has been proofread and is free of spelling and grammatical errors.	All of the following: • The Reporting "C" criteria; • Regular, well-commented and incremental GitHub commi	All of the following: • Submission before the deadline; • First-Walk-and-Talk milestone submitted; • GitHub utilised; • Post-Mortem PDF submitted.	Any one of the following: • No coversheet included; • No First-Walk and-Talk milestone submitted; • No Post-Mortem PDF submitted.